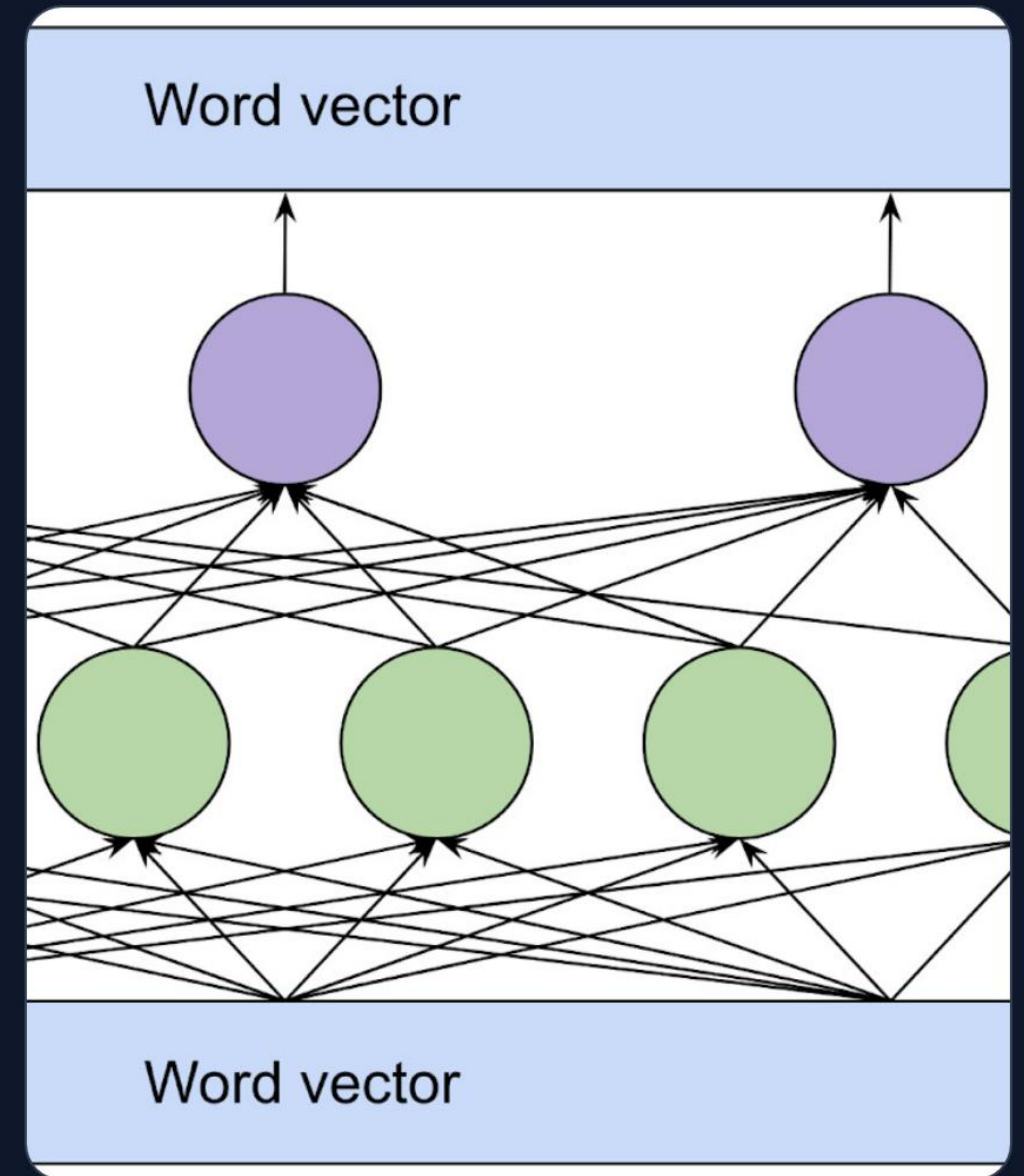


LLM ARCHITECTURE & MECHANISM OVERVIEW

Inside the LLM: Text, Vectors & Context

How Transformer neural networks turn raw words into high-dimensional space and resolve ambiguous meanings dynamically.



The Four-Stage Text-to-Vector Journey

STAGE01

Tokenization

Splitting raw string inputs into discrete vocabulary IDs.



STAGE02

Static Embedding

Mapping IDs to initial high-dimensional coordinate vectors.



STAGE03

Self-Attention

Blending surrounding token context into dynamic representations.



STAGE04

Softmax Sampling

Projecting vector hidden states to next-token probabilities.

Tokenization: From Words to Vocabulary IDs

RAW INPUT STRING

“The salmon swam near the river bank.”



SUBWORD CHUNKS (BPETOKENIZER)

[The]

[salmon]

[swam]

[near]

[the]

[river]

[bank]



NUMERICAL TOKENIDS (VOCABULARY LOOKUP)

[464, 21873, 28104, 2266, 262, 2762, 1842]

Static Token Embeddings: The Geometry of Words

Lookup table converts discrete integer IDs into continuous d-dimensional vectors.

EMBEDDING MATRIX LOOKUP

Token ID: 1842 ("bank")

↓ W_embed Matrix Lookup

`[-0.14, 0.82, -0.45, 0.29, ...]`



High Dimensions

768 to 12,288 floating point numbers per vector.



Positional Encoding

Adding sequence index vectors to preserve word order.

THE POLYSEMY DILEMMA

Why Static Embeddings Fail Alone

The same word token possesses completely different meanings depending on context.

CONTEXT A • GEOGRAPHY

“The salmon swam near the river bank.”

Meaning: Natural river boundary / earth slope

STATIC LOOKUP

ID: 1842



SAME VECTOR

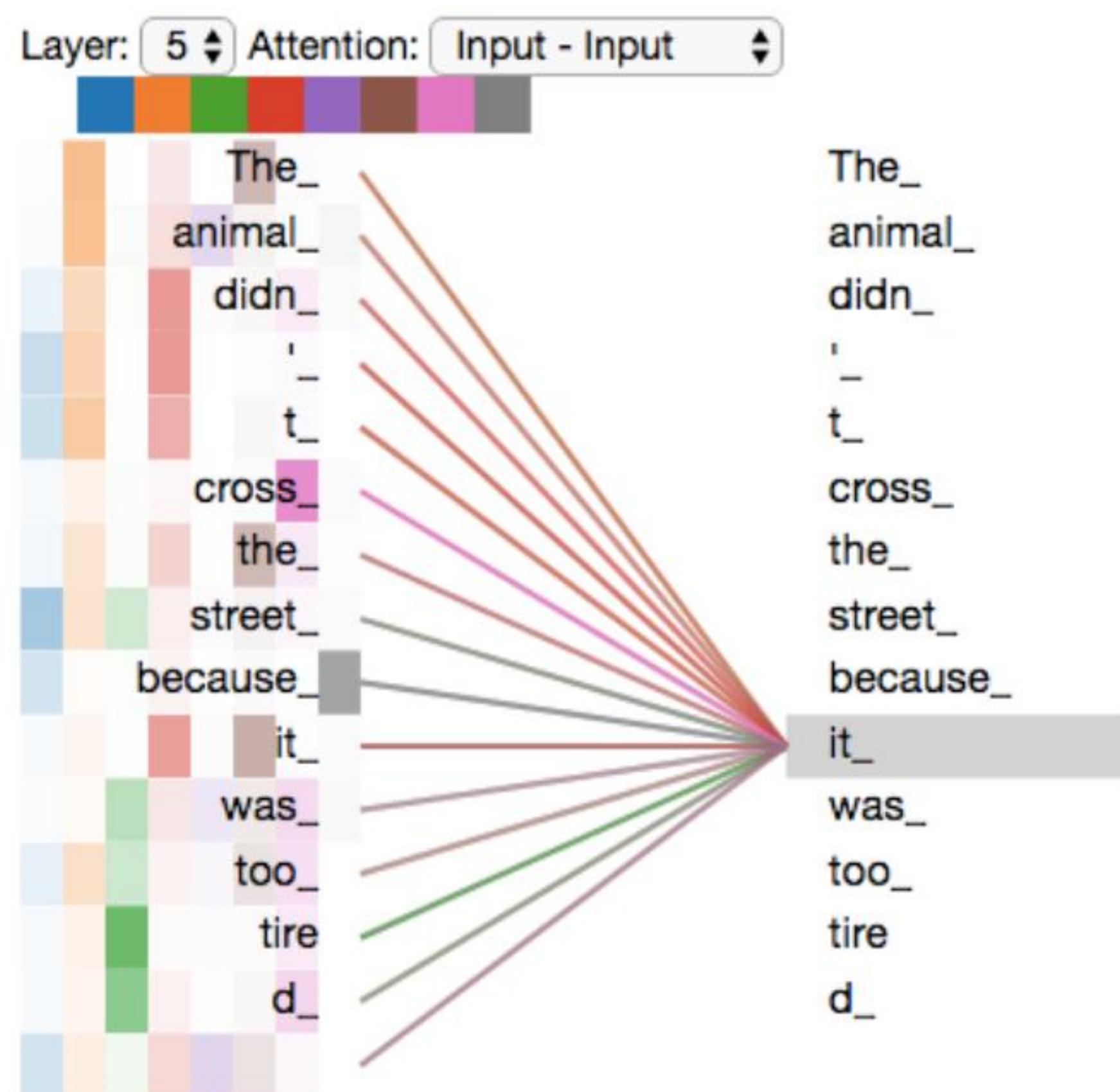
CONTEXT B • FINANCE

“The investor deposited cash at the bank.”

Meaning: Financial institution / vault

Self-Attention: Routing Information Between Tokens

Query, Key, and Value projections allow tokens to query surrounding neighbors.



Query (Q)

“What contextual information am I seeking?”

Key (K)

“What semantic features do I offer?”

Value (V)

“What exact information do I share?”

How Context Repositions the "Bank" Vector

Dot-product attention scores shift the vector position in internal state space.

SCENARIO A • RIVER BANK

Prompt: "The water flowed along the bank ."

ATTENTION WEIGHTS TO "BANK"

water	0.68
flowed	0.22
along	0.10

Shifted toward Hydrology Cluster

SCENARIO B • MONEY BANK

Prompt: "He deposited money at the bank ."

ATTENTION WEIGHTS TO "BANK"

deposited	0.64
money	0.29
at	0.07

Shifted toward Finance Cluster

Layer-by-Layer Refinement in Residual Streams

Vectors pass through dozens of stacked attention and MLP layers.

EARLY LAYERS (1–8)

Syntax & Grammar

Parsing local word relationships and parts of speech.



MIDDLE LAYERS (9–24)

Context Resolution

Disambiguating polysemous words via self-attention.



LATE LAYERS (25–32+)

Next-Token Target

Preparing final hidden state for output projection.

From Vectors to Next-Token Probabilities

Unembedding matrix projects hidden states to vocabulary logit scores.

Candidate Token

Softmax Probability

"water"

74.2%

"mud"

12.5%

"money"

0.8%

Sampling Parameters

Temperature: Controls randomness of probability distribution curve.

Top-P (Nucleus): Restricts selection to top cumulative probability mass.

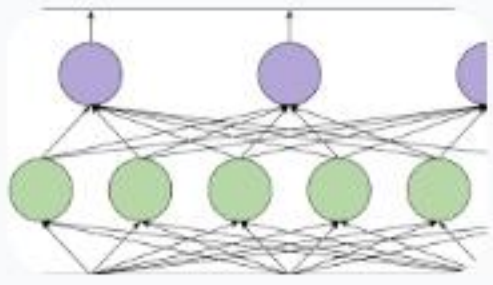


Key Takeaway

LLMs do not store static word meanings. They operate in high-dimensional vector spaces, using self-attention to dynamically synthesize context and predict the most coherent next token.

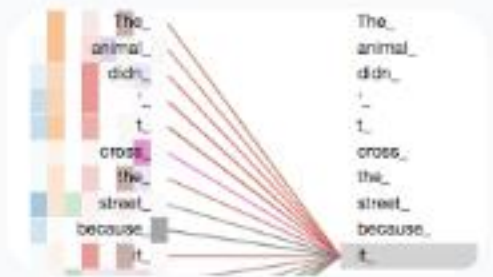


Image Sources



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Source: www.understandingai.org



https://jalammar.github.io/images/t/transformer_self-attention_visualization_3.png

Source: jalammar.github.io